

Causal ML: shrinkage and identification

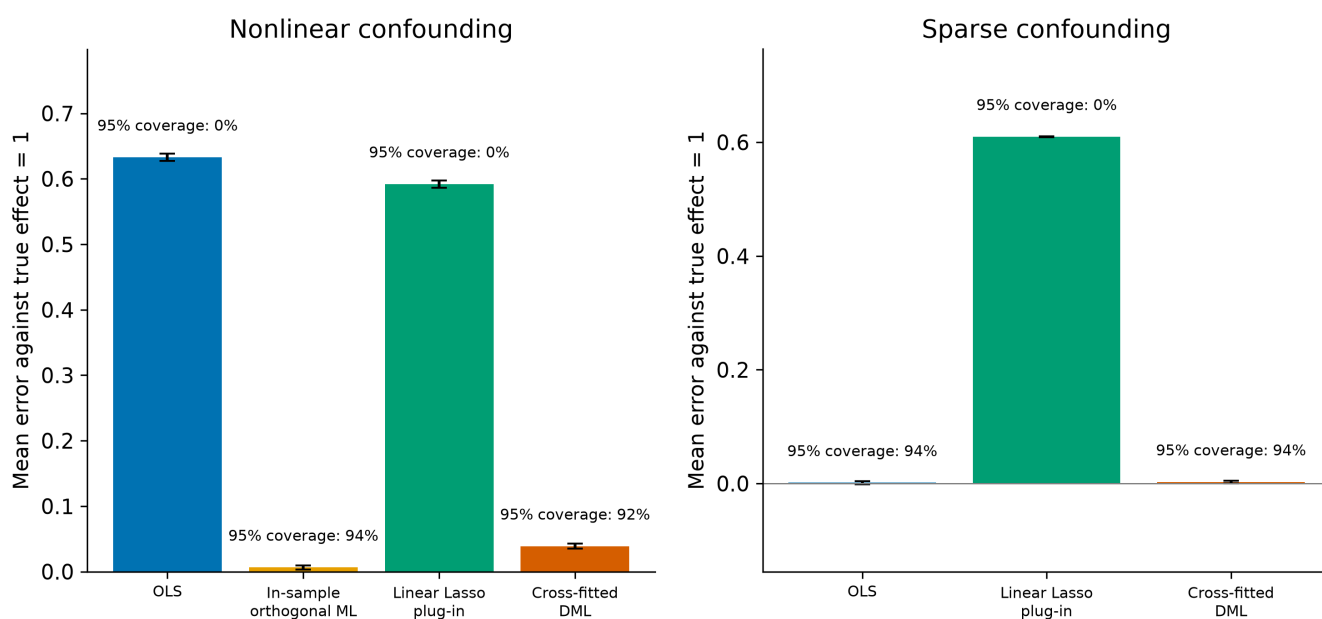
Hakan Zeki Gulmez | Research brief | 7 October 2026

Question

When can regularization distort a treatment effect, and what assumptions remain?

Method

We compare unchanged estimators in two fixed simulations with known effects. We increase Monte Carlo precision and add overlap and confounding-sensitivity diagnostics to the historical SIPP replication. We retain the original staggered-adoption analysis and add a narrowly supported, balanced late-cohort window.



1000 repetitions per design; bars: 95% Monte Carlo intervals for mean bias.
Coverage describes effect intervals. Nonlinear design: boosting nuisances; sparse design: Lasso nuisances.

Sources: fixed seeded experiments; private public-SIPP/MPDTA retention. Aggregate V3 CSVs and inherited pension.csv.

Three findings for the decision-maker

Hakan Zeki Gulmez | Research brief | 7 October 2026

1. In 1,000 sparse-design repetitions, linear Lasso plug-in bias is 0.610, versus 0.003 for cross-fitted DML; nominal 95% coverage is 0.0% versus 93.8%.
2. The nonlinear design qualifies that message: in-sample orthogonal ML bias is 0.007, versus 0.039 for cross-fitted DML. These fixed learner/design comparisons do not establish universal DML superiority.
3. The historical RF-IRM estimate remains \$8,172. No observations are trimmed; sensitivity RV is 8.41% and RVa 4.94% under the specified adversarial confounding model, not proof of no confounding.

Limitations

- Designs and learner families differ; rankings are not universal.
- IPW ESS measures weight concentration, not causal validity or propensity-estimation uncertainty.
- Historical eligibility is not randomized or modern policy participation.
- Balanced and full-sample event studies target different cohorts/windows.
- Effect and group inference conditions on fitted nuisance structures.

Detailed assumptions, diagnostics and source lineage: README Technical notes and report/report.md.